



A 5G-TSN joint resource scheduling algorithm based on optimized deep reinforcement learning model for industrial networks

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ABSTRACT

As the Industrial Internet of Things (IIoT) evolves, the rapid growth of connected devices in industrial networks generates massive amounts of data. These transmissions impose stringent requirements on network communications, including reliable bounded latency and high throughput. To address these challenges, the integration of the fifth-generation (5G) mobile cellular networks and Time-Sensitive Networking (TSN) has emerged as a prominent solution for scheduling diverse traffic flows. While Deep Reinforcement Learning (DRL) algorithms have been widely employed to tackle scheduling issues within the 5G-TSN architecture, existing approaches often neglect throughput optimization in multi-user scenarios and the impact of Channel Quality Indicators (CQI) on resource allocation. To overcome these limitations, this study introduces ME-DDPG, a novel joint resource scheduling algorithm. ME-DDPG extends the Deep Deterministic Policy Gradient (DDPG) model by embedding a Modulation and Coding Scheme (MCS)-based priority scheme. This improvement in computational efficiency is critical for real-time scheduling in IIoT environments. Specifically, ME-DDPG provides latency guarantees for time-triggered applications, ensures throughput for video applications, and maximizes overall system throughput across 5 G and TSN domains. Simulation results demonstrate that the proposed ME-DDPG achieves 100 % latency reliability for time-triggered flows and improves system throughput by 10.84 % over existing algorithms under varying Gate Control List (GCL) configurations and user ratios. Furthermore, due to the combination of MCS-based resource allocation scheme with DDPG model, the proposed ME-DDPG achieves faster convergence speed of the reward function compared to the original DDPG method.

1. Introduction

IIoT constitutes a crucial technological foundation for promoting automation, enhancing intelligence within industrial production and realizing smart manufacturing [1–4]. In modern manufacturing environments, numerous interconnected devices and systems, including sensors, machinery, and control systems, rely on robust and efficient communication networks [5–7]. To ensure seamless interoperability among these heterogeneous devices, industrial networks are required to support diverse communication protocols and standards. This capability is vital for real-time data acquisition, monitoring, and control, thereby laying the groundwork for optimizing production processes, enhancing operational efficiency, and ensuring safety [8]. With the integration of IIoT and artificial intelligence (AI), the demand for reliable, low-latency, and high-throughput communication has become more prominent [9,

10]. The combination of these technologies generates and processes larger volumes of data, demanding more advanced network infrastructure to handle various communication tasks. Hence, the development and implementation of advanced industrial networks are essential for achieving the goals of smart manufacturing, driving innovation, and maintaining competitive edge in the industrial sector [11].

In industry scenarios, common data flows consist of time-triggered flows and video flows. The time-triggered flow holds a higher priority as it encompasses the information of control tasks of production equipment, such as the action instructions of mechanical arm and motor. This type of service is typically periodic, and imposes high requirements on transmission performance metrics such as latency, jitter and reliability [12]. Specially, in the industrial low-latency scenario, the latency between end-to-end devices is constrained to the millisecond range, while jitter is kept within the microsecond range, and the transmission

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success rate is required to reach 99.9999 %. In contrast, video flows, sampled by collecting devices, are used to facilitate intelligent decision-making in various IIoT-related businesses like target detection, fault diagnosis, and remote monitoring. High-speed sampling causes video flows to consume more bandwidth resources, thus necessitating the network to provide throughput guarantees [11].

To address the challenge of deterministic communication in industrial networks, the combination of 5 G and TSN has emerged as a solution. 5 G provides large capacity, high speed, low latency, and high mobility. Specially, the 5 G system (5GS) is capable of offering over one million connections per square kilometer, thereby laying a solid foundation for IIoT transmission [13,14]. TSN, which is standardized by the IEEE 802.1 task group, represents an Ethernet-based networking technology. It can uniformly carry multiple services on the Ethernet physical layer, thus providing bounded latency and highly reliable transmission for high-priority industrial control services. With the deepening development of the industrial Internet, the convergence of information technology (ICT) and operational technology (OT) has been accelerated. TSN has emerged as a new industrial network technology that furnishes a unified and open network architecture for the construction of intelligent factories. It is compatible with various industrial network protocols and guarantees the unified support and reliable performance of IT operations within factories. [15]. Moreover, TSN enables time-critical applications to be transmitted at precise intervals. The time-triggered scheduling paradigm ensures bounded latency, even when multiple flows simultaneously compete for the same network resources [16].

The 3GPP has incorporated TSN into 5GS since Release 16 [17]. Through the integration of 5G-TSN and by capitalizing on the characteristics of two common data streams in industrial networks, the challenge of attaining deterministic end-to-end industrial communication can be addressed. This is achieved by guaranteeing latency for time-triggered flows, ensuring throughput for video flows, and maximizing the overall throughput. According to 3GPP specifications, the 5GS architecture has been enhanced with a series of new functionalities to seamlessly integrate TSN and 5 G in both the control and data planes. However, it should be noted that 5 G and TSN are based on fundamentally different networking principles, such as differences in physical layer techniques, MAC (Media Access Control) schemes, and upper-layer protocols. Therefore, to ensure deterministic communication between 5 G and TSN, it is crucial to make dynamic resource allocation decisions at the 5 G base station (gNB) based on the GCL status [18,19].

2. Related works

2.1. Overview of TSN scheduling mechanisms

The IEEE 802.1 TSN task group has developed a series of standards to enable Ethernet-based networks to provide reliable real-time communication for both time-critical and non-time-critical flows. These standards cover aspects such as time synchronization, traffic scheduling, and resource management, thereby facilitating the delivery of deterministic services within local area networks (LANs).

In TSN, scheduling mechanisms are essential for ensuring bounded latency. The primary methods are described as follows:

Time-Aware Shaping (TAS): Specified in IEEE 802.1Qbv. TAS is a time-triggered traffic scheduling mechanism ensuring deterministic guarantee for real-time flows. Firstly, it maps flows with different QoS requirements into various priority queues at the egress port of a TSN switch. Secondly, as illustrated in Fig. 1, TAS utilizes synchronized clocks to regulate the transmission periods of each queue, enabling precise control of the timing and volume of frame transmissions. Each priority queue is controlled by a gate mechanism, which periodically opens or closes based on a pre-configured time list called GCL. Based on GCL, packets in the high-priority queue are transmitted without interruption from low-priority queues, ensuring that high-priority traffic obtains guaranteed bandwidth and avoids queuing delays, making TAS particularly suitable for industrial applications requiring strict timing guarantees.

Cyclic Queueing and Forwarding (CQF): Specified in IEEE 802.1Qch. Although TAS can achieve “zero-jitter” and “no waiting scheduling” for hard real time flows, the GCL planning for large scale switching nodes and massive flows is too complex. CQF proposes a deterministic scheduling method that is easy to implement regardless of network topology. CQF divides the time into time slots with the same duration T , named as cycle, meanwhile, two queues are deployed for packet receiving and transmission. In one cycle, one queue is responsible for packet receiving but closed for packet transmission while another queue is only permitted for packet transmission. In the next cycle, the responsibilities of two queues are swapped, with the previous receiving queue allowed to transmit packets received in the last cycle and the previous transmission queue changed to receive packets. Following this manner, CQF can assure that the packet received in current cycle would be forwarded to the next node in the next cycle. However, the jitter of flows adopting CQF is from 0 to $2T$, which cannot provide absolutely no waiting scheduling as TAS. Furthermore, similar to TAS, CQF also needs time synchronization among nodes.

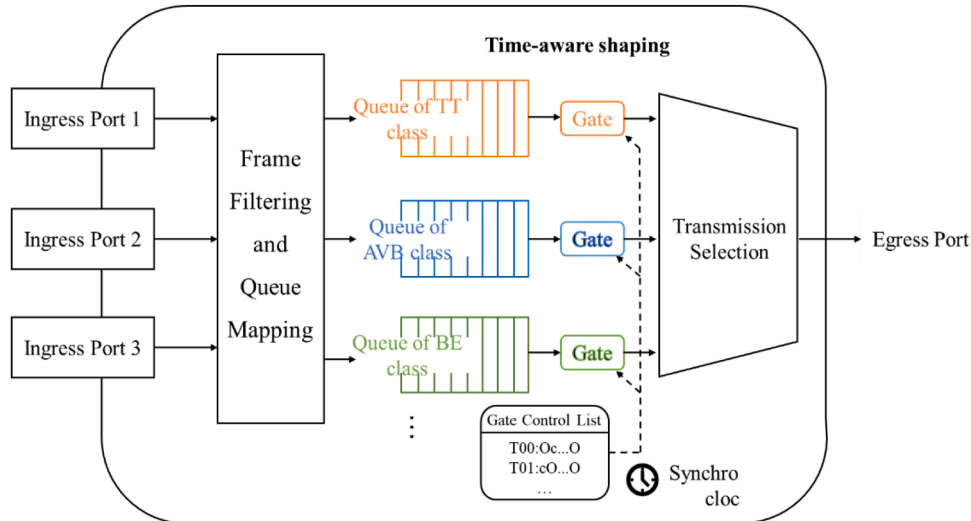


Fig. 1. Time Aware Shaping in TSN Switch.

Asynchronous Traffic Shaper (ATS): Specified in IEEE 802.1Qcr. Unlike TAS and CQF, ATS doesn't need time synchronization among network nodes, which is suitable for large scale deterministic communications. ATS is a flexible and efficient traffic shaping mechanism designed to handle diverse traffic patterns. Unlike traditional shaping mechanisms, ATS calculates an eligibility time for each packet upon arrival. This eligibility time is related to the maximum allowable residence time in one switching nodes and determines when a packet can be transmitted, allowing the scheduler to avoid real-time recalculations during the transmission phase. Additionally, ATS enables hop-by-hop shaping, prevents the accumulation and amplification of bursts across the network, and improves overall performance. ATS is particularly suitable for time-sensitive networking scenarios with mixed-criticality traffic, such as industrial automation, real-time control systems, and audio-video streaming, where low latency, low jitter, and deterministic behavior are critical. Due to the lack of a time synchronization mechanism in ATS, it is relatively weak in terms of delay determinism and it is difficult to ensure stable low-latency services in various situations.

Therefore, among these mechanisms, TAS is particularly suitable for scenarios requiring strict timing guarantees, such as industrial automation [20]. The pre-configured and cyclically operated GCL determines the network resources available for each queue, effectively managing transmission rates and ensuring that time-critical applications receive the necessary bandwidth and priority. Through the implementation of TAS, TSN ensures that the network can meet the stringent requirements of industrial automation.

2.2. 5G-TSN integration architecture

The integration of TSN protocols into 5 G networks of great significance for supporting industrial applications with requirements of strict timing and low-latency communication. As shown in Fig. 2, specific functional entities are added to 5GS in both the control and data planes, thereby transforming the 5GS into a TSN bridge. This section focuses on the essential components and their roles in the 5G-TSN integrated architecture, with an emphasis on the coordination necessary for efficient joint scheduling.

In the TSN domain, there are two critical entities that facilitate service management and network control: the Centralized User Configuration (CUC) and the Centralized Network Configuration (CNC). The CUC undertakes the tasks of collecting service requirements and identifying TSN capabilities at end stations. It acts as an intermediary between user requirements and network capabilities. Meanwhile, the CNC functions as the network controller, managing network topology discovery, transmission path selection, and GCL orchestration. To achieve seamless compatibility with the CNC, namely the TSN Application Function (TSN-AF), is introduced in the 5 G control plane. Adhering to

IEEE 802.1Qcc standards, the TSN-AF conveys the status of the 5G-TSN bridge to the CNC and receives critical service information, including flow periods, directions, packet specifications, and QoS requirements for time-sensitive applications. Subsequently, the TSN-AF relays this QoS information to the Policy Control Function (PCF) within the 5 G core network, facilitating the establishment of appropriate QoS profiles. In the data plane, the Network-side TSN Translator (NW-TT) and the Device-side TSN Translator (DS-TT) function as gateways between the TSN and 5 G domains. These translators are in charge of protocol translation and data forwarding. Both NW-TT and DS-TT monitor the status of time-sensitive applications, report to the TSN-AF, and obtain configuration details from the CNC through the TSN-AF. To manage flows and schedule resources, TAS is implemented within NW-TT and DS-TT. This enables data forwarding with precise timing and effectively reduces delay for time-sensitive applications. The scheduling capabilities provided by TAS are vital for maintaining the synchronization and timing required in industrial applications. Through the integration of these components and functionalities, the 5G-TSN architecture constructs a robust framework with the capability of scheduling multiple services.

2.3. 5G-TSN joint scheduling

Under the 5G-TSN integration architecture, end-to-end scheduling mechanisms are investigated to guarantee deterministic communications for time-triggered traffics. However, when scheduling involves both 5 G and TSN domains, significant challenges arise. The fundamental differences between 5 G and TSN, including their physical layer technologies and MAC protocols, contribute to the difficulty. Hence, 5G-TSN joint scheduling is a cross-domain issue that requires the coordination of resources in different network domains. Orthogonal Frequency Division Multiplexing (OFDM) is a key technology in 5 G networks, known for its efficient use of spectrum utilization and interference resistance. In OFDM, the available bandwidth is divided into multiple orthogonal sub-carriers that can be independently modulated and demodulated, enabling high data rates and flexible resource allocation.

In the context of 5 G, the smallest unit of resource allocation is the Resource Block (RB). RBs are allocated to users in both the time and frequency domains, providing the granularity necessary to manage the diverse requirements of different applications and services. Each RB consists of a fixed number of sub-carriers over a specified number of OFDM symbols, ensuring efficient and flexible use of the available spectrum. This modular approach facilitates dynamic resource management and enhances the overall network performance. Therefore, scheduling in 5 G focuses on the RB allocation capability of gNB at the air interface, while TSN emphasizes designing GCL configurations and scheduling slots for different flows.

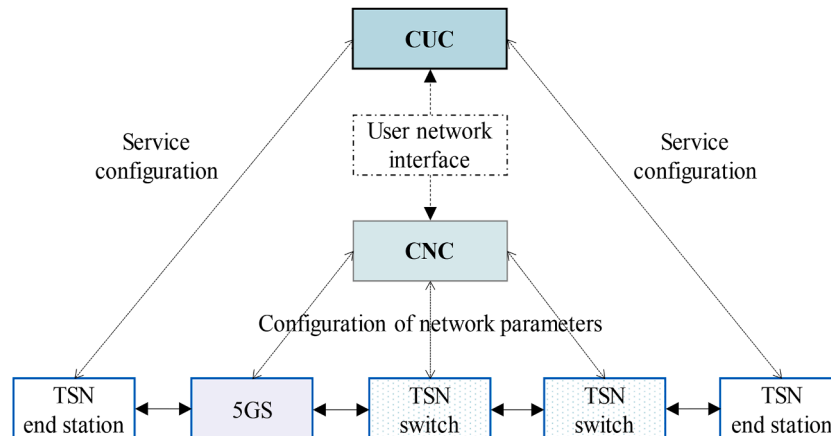


Fig. 2. 5G System as a TSN Bridge.

There are several existing studies investigating scheduling problems in 5G-TSN architecture. In [21], Martenvormfelde et al. presented a simulation model using OMNet++, detailing the transmission procedures in a 5G-TSN environment. However, this study does not propose any specific scheduling schemes. In [22], Ginthör et al. logically divided the SDN scheduler into 5G-side scheduler, TSN-side scheduler and end-to-end scheduler. The scheduling algorithm under this framework could achieve high resource utilization. In [23], Zhang et al. integrated Virtual Network Function (VNF) into 5G-TSN framework and designed a mixed-integer linear programming resource scheduling method, which realized the decoupling of infrastructure and application functions, satisfying the diversified QoS requirements of industrial applications and achieving good performance in terms of delay and success rate. In [24], Cai et al. proposed a dynamic QoS mapping method based on the improved K-means clustering algorithm and rough set theory to ensure the deterministic transmission of time-sensitive flows in the 5G-TSN network. Nazari et al. in [25] proposed a novel Incremental Joint Scheduling and Routing (IJSR) framework for 5G-TSN integration, offering a heuristic solution for deterministic communication. However, those methods discussed above are difficult to apply to the large-scale and dynamic networks.

Currently, reinforcement learning (RL) has gained widespread attention as an effective approach to address scheduling problems [26]. RL can effectively model the relationship between the state and action. Wang et al. in [27] proposed a hierarchical particle swarm optimization algorithm (DQHPSO) with double Q-learning to find optimal solutions for 5G-TSN joint scheduling. However, it only focuses on time-triggered flows and does not consider other types of services. The work by Yang et al. in [28] developed a GCN-based deep reinforcement learning (DRL) method to solve the joint optimization target, but it didn't consider the 5 G domain and would face action space aggregation of DRL in large-scale networks. In [29], Prados-Garzon et al. addressed the online flow allocation problem (OFAP) in ATS-based asynchronous networks. It presented a RL solution that combines data-driven techniques with analytical models to enhance the network operator's revenue. Li et al. investigated the co-design of flow scheduling in 5 G and TSN [30], implementing 5G's grant-free configured-grants (CGs) and TSN's CQF mechanisms. In [31], Zhu et al. proposed a DDPG algorithm, which delayed the transmission of time-triggered flow as much as possible and encouraged the transmission of video flows to improve the throughput of the 5G-TSN system. However, it fails to address scenarios where time-triggered flows have sufficient resources, do not need to be postponed the scheduling, and it also lacks a detailed allocation method for user scheduling. In [32], Feng et al. proposed a dynamic scheduling method based on a Delay-based Proportional Fairness (DPF) algorithm to balance low latency and fairness in resource allocation, but the improvement of the throughput by this algorithm is not significant.

Based on the current state-of-the-art on 5G-TSN joint resource scheduling, it can be observed that joint scheduling is traditionally modeled as a non-deterministic polynomial (NP-hard) problem, then transformed into optimization problems and solved by heuristic algorithms or intelligent algorithms. However, due to the complexity of nonlinear objective functions and multiple optimization factors, obtaining analytical solutions is challenging, and the design of numerical iterative algorithms significantly affects the resource scheduling performance. The methods discussed in [26–32] cannot simultaneously consider large-scale networks, maximum throughput, lower action space, and dynamic CQI. This article proposes an adaptive 5G-TSN joint scheduling method using an DDPG algorithm embedded a priority scheme. The proposed method considers the GCL setting in the TSN domain, the characteristics of multiple applications, the number of available RBs, and the current CQI in the 5 G domain. The optimization objectives are to reliably guarantee the latency requirements of time-critical applications, to secure the minimum throughput for video applications, and to maximize the overall throughput for all applications. Priorities are differentiated based on the characteristics of data

flows and CQI. Specific RBs are then identified after the acceptance of the binary output generated by the DDPG algorithm. This approach utilizes the non-Markov CQI and reduces the action space of the DDPG algorithm. Specifically, this paper processes the multi-dimensional information of the dual domains, thereby highly reliably meeting the low-latency requirements of time-sensitive services and enhancing the overall throughput of the system.

3. System model and problem formulation

In 5G-TSN industrial networks, multiple applications coexist and transmit simultaneously. High-priority data streams, typically related to critical tasks and real-time control applications, are known as time-triggered flows. Medium-priority data streams may include diagnostics, condition monitoring, and applications like remote surveillance and visual inspection. Video flows are considered medium priority due to their real-time transmission requirements and higher bandwidth demands. Our system model integrates both 5 G and TSN to form the industrial network. Applications within this network are classified into two main types: time-triggered and video applications. It is assumed that 5G-TSN network consists of M TSN users and N video users, denoted as two different flow sets F_t and F_v . For the ease of analysis, we assume that each terminal receives only one type of flow. The set of all flows is presented as $F, F = F_t \cup F_v$. The currently scheduled flow is denoted as $i, \forall i \in F$. In a 5G-TSN industrial network, time-triggered flows demand stringent end-to-end latency limits, while video flows require minimum throughput guarantees. The definitions and specific QoS requirements for these applications are detailed below:

$$F_t = \{f_{t,1}, f_{t,2}, \dots, f_{t,M}\} \quad (1)$$

$$F_v = \{f_{v,1}, f_{v,2}, \dots, f_{v,N}\} \quad (2)$$

$$d_i < d_{qos} \quad \forall i \in F_t \quad (3)$$

$$\bar{th}_i \geq th_{min} \quad \forall i \in F_v \quad (4)$$

where d_i is the maximum end-to-end delay of flow i in F_t in the 5 G system, and d_{qos} is the latency requirement of F_t ; $\bar{th}_i$ is the achieved average throughput of the scheduled flow i in F_v , and th_{min} is the minimal throughput requirement of F_v .

In 5 G communication systems, for the reliable transmission of time-sensitive data, the adaptive MCS is adopted. MCS maps different Signal-to-Interference plus Noise Ratio (SINR) to corresponding values, which combine modulation methods and the coding rate of Forward Error Correction (FEC) in physical layer. Common modulation methods include Binary Phase Shift Keying (BPSK) with strong anti-interference ability but a low data rate using two phases for modulation, Quadrature Phase Shift Keying (QPSK) balancing anti-interference ability and data rate using four phases, 16-level Quadrature Amplitude Modulation (16-QAM) providing a higher data rate under higher SINR conditions by combining amplitude and phase, and 64-level Quadrature Amplitude Modulation (64-QAM) achieving higher data transmission rates with 64 different amplitude and phase combinations when the SINR is high.

When transmitting data packets over a wireless channel, setting MCS based on CQI can enhance resistance to interference and signal attenuation. Our model adopts adaptive MCS by translating the SINR to the required RB numbers for flow i in each Time Transmission Interval (TTI). The mapping relationship from SINR to bytes permitted in one RB is shown in Table 1. The values in the table are set according to the rules defined in 3GPP, higher SINR corresponds to higher modulation order.

Then, the scheduled throughput of flow i is calculated as:

$$\bar{th}_i = \frac{\sum_{t=1}^T RB_{scheduled,i}^t * b_i^t}{T} \quad (5)$$

Table 1
Relationship of SINR, Modulation Method, MCS and Permitted Bytes per RB.

SINR Range (dB)	Modulation Method	Bits withing related with MCS	Permitted Bytes per RB
< 10	QPSK	4	84
10–20	16-QAM	12	252
> 20	64-QAM	20	420

Assuming scheduling begins from the 1st TTI, where each TTI has a period of T , $RB_{scheduled,i}^t$ denotes the actual number of RBs scheduled for flow i , and b_i^t represents the number of bytes per RB according to the relevant MCS. Thus, it can be inferred that with the variation of the CQI at each moment, b_i^t corresponding to each data stream will change when crossing the critical values, and the corresponding throughput will change accordingly. To calculate the required number of RBs in each queue, the bits permitted in one RB are firstly calculated as:

$$m_i = B \log_2(1 + g_i) - Q^{-1}(\varepsilon) \sqrt{V(g_i)} \quad (6)$$

where ε is the tolerable bit error rate, B is the frequency bandwidth of one RB, $Q^{-1}(\varepsilon)$ and $V(g_i)$ are defined as:

$$Q^{-1}(\varepsilon) = \frac{1}{\sqrt{2\pi}} \int_{\varepsilon}^{\infty} e^{-\frac{t^2}{2}} dt \quad (7)$$

$$V(g_i) = (\log_2(e))^2 \left(1 - \frac{1}{(1 + g_i)^2} \right) \quad (8)$$

Then the required numbers of RB in flow i can be obtained from:

$$RB_{required,i}^t = \frac{s_i \times l_i}{m_i} \quad \forall i \in \mathbf{F} \quad (9)$$

where s_i and l_i mean the packet size and packet number in queue of flow i .

As shown in Fig. 3, the process for 5 G and TSN joint scheduling for downlink data transmission is based on an AI agent. This agent functions as a logical entity within the gNB and is responsible for flexible resource scheduling decisions. The agent learns from the environment to determine the appropriate “action”. In this process, F_t are scheduled by gNB

and transmitted sequentially through DS-TT and TSN User Equipment (TSN-UE). F_t are scheduled by gNB and directly delivered to the Video User Equipment (Video-UE). Packets of F_t are mapped into the GCL, which is pre-defined in DS-TT. The transmission duration for the time-triggered queue is 1 TTI, indicating that the ON/OFF state alternates in each TTI. The detailed scheduling procedures are as follows:

Step 1: At the start of a TTI, the agent gathers environment information, including the GCL state of the time-triggered queue in DS-TT and the status of each active flow’s queue in gNB, such as the number of queued packets and the queuing delay for each flow. The agent is physically located in the gNB and receives the queue state from the gNB via an internal interface, while the CQI and GCL state are reported by the UE through radio signaling. Additionally, each user generates a variable number of data packets, causing the number of queued packets and queuing delay for each flow to change in each TTI.

Step 2: All application packets are buffered in the gNB for downlink scheduling. Using information gathered from UEs, gNB, and DS-TT, the agent decides which flow to schedule based on the number of packets awaiting transmission, service priority, obtained $\bar{th}_i$, and d_{qos} , and selects which flow can be scheduled. Policy-Net of the DDPG agent outputs binary scheduling actions A_t^i determining which user will be allocated RBs. The number of RBs is finite, and the sum of allocated RBs should not exceed the available RBs. When the following condition is satisfied:

$$\sum_{i=1}^k A_i^t * RB_{required,i}^t \leq RB_{available} \quad (10)$$

where A_i^t is the binary output of DDPG, $RB_{required,i}^t$ indicates the required RB numbers of flow i calculated by Eq. (9), $RB_{available}$ is the number of available RBs. If Eq. (10) is not satisfied, the number of RBs is insufficient to allocate, it means that $RB_{scheduled,i}^t \leq RB_{required,i}^t$ under $A_i^t = 1$ condition, and the scheduling process moves to Step 3. If Eq. (10) is satisfied, RBs allocated by the environment are sufficient to allocate, and $RB_{scheduled,i}^t$ can be assigned as:

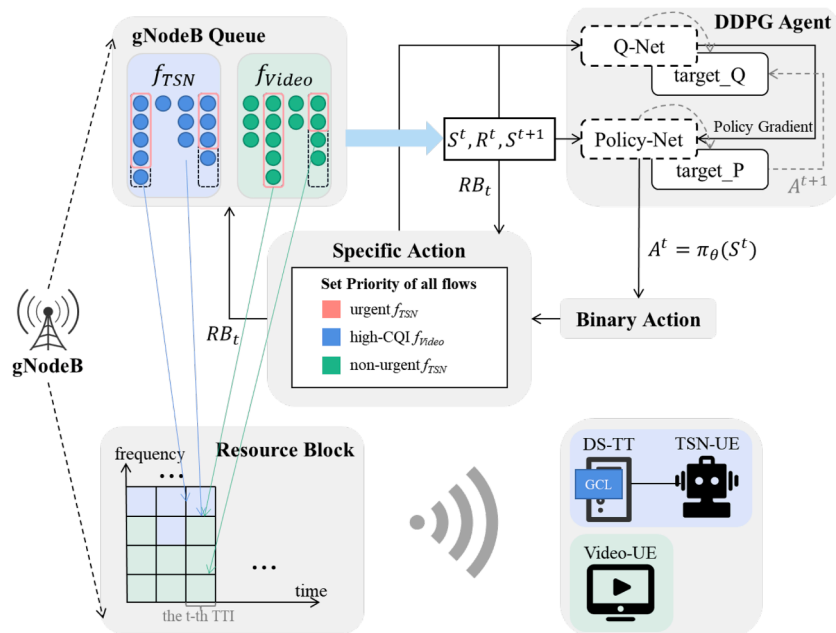


Fig. 3. Scheduling Process.

$$RB_{scheduled,i}^t = \begin{cases} RB_{required,i}^t & \text{if } A_i^t = 1 \\ 0 & \text{if } A_i^t = 0 \end{cases} \quad (11)$$

Step 3: MCS-based priority calculation and scheduling decision-making. In current TTI, flows are divided into three types: urgent F_b , F_v , and non-urgent F_t . Within each type of data flow, RBs are allocated in order of descending MCS levels. For the queues that executed "1" in Step 2, RBs are first allocated to F_t whose latencies are close to QoS requirements, referred to urgent F_t . The remaining RBs are then distributed to F_v based on their MCS levels. When the MCS levels of data packets are identical, RBs are allocated based on queue length. If any RBs remain not allocated, they are then assigned to non-urgent F_t in order of MCS levels. Once the number of RBs to be allocated is determined, transmission can proceed accordingly, gNB transmits data packets to UEs, and the allocated resources are configured for packet delivery of each eligible flow.

Once the agent allocates radio resources to the corresponding flow, the environment changes, including the GCL state in DS-TT, the queue state in gNB, and the channel state in the UE. The agent then learns new states and makes the suitable actions for the next TTI. This means that the Step 1 to 3 will be performed periodically in each TTI. As previously discussed, the agent functions as a scheduler to balance QoS between F_t and F_v through joint optimization of radio resource allocation. The scheduling objective in this article is to ensure solid latency guarantees for F_b , minimum throughput guarantees for F_v , and maximum total throughput. To achieve these goals, the utility function of flow i at the t^{th} TTI is defined as:

$$U_i(t) = \begin{cases} (d_{qos} - d_i - d_{GCL}) * (\bar{th}_i - th_{min}) & \text{if } d_i \leq d_{qos} \text{ and } \bar{th}_i \geq th_{min} \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

where d_{GCL} represents the delay caused by GCL.

$$d_{GCL} = \begin{cases} 1 & i \in F_t \cap GCL(t) = 0 \\ 0 & i \in F_t \cap GCL(t) = 1 \text{ or } \forall i \in F_v \end{cases} \quad (13)$$

where $GCL(t)$ represents the GCL state in TSN domain, $GCL(t) = 1$ means that the GATE is open in current TTI in TSN, otherwise $GCL(t) = 0$. d_{GCL} is set to 1 only when the active flow is belonged to F_t and the GATE is closed. The calculation of $\bar{th}_i$ is according to Eq. (5).

In Eq. (12), since the packet size of F_t is smaller than that of the F_v , priority transmission of F_v to should be encouraged to enhance the throughput under the strict delay guarantee of F_t . Therefore, the value of $U_i(t)$ is higher when $(d_{qos} - d_i - d_{GCL})$ is aim to minimize the difference between d_{qos} and d_i , and considering the GCL state of TSN. $U_i(t)$ is positively correlated with $\bar{th}_i$ of F_v as well. Assuming the system has K users to allocate RBs in current TTI, the discussed issue can be transformed as an optimization problem:

$$\max \sum_{i=1}^K U_i(t) \quad (14)$$

subject to Eqs. (3), (4), (10) and (13).

The above optimization problem is non-convex and involves both continuous and discrete variables. In this case, the relationship between the observation and resource allocation scheme is implicit, thus it is hard to analytically solve the non-convex problem. But DRL model can find the relationship and optimize it. However, concerning the optimization problem discussed in this paper, it may lead to massive action space for DRL model. If the allocated RBs for each scheduled flows are directly set as the model's output, it will impose a heavy computational consumption for searching the optimal. To address this issue, a priority scheme with low computational complexity is introduced into the training loop

of DRL model to replace its continuous action space.

4. Resource scheduling algorithm based on ME-DDPG model

4.1. Markov model of DDPG

In RL, the fundamental aim is to seek an optimal policy which maximizes the cumulative rewards obtained from the environment [33]. DRL analyzes environmental data for the purpose of making decisions and attaining desired rewards. Deep Q-learning conventionally involves multiple dimensions of Q-values corresponding to each action. In contrast, DDPG, which stem from the Actor-Critic framework, is designed to improve policy optimization efficiency and computational speed over that of standard DRL. The actor-network in DDPG learns a deterministic policy, enabling it to map extensive states to actions, thereby facilitating fine-grained control outputs. A DDPG-based resource scheduling algorithm is developed to ascertain the optimal result, rendering it well-suited for tasks involving multiple users. The Markov Decision Process (MDP) supplies the mathematical structure for DDPG, consisting of state space $\mathcal{S}$, action space $\mathcal{A}$, transition probability P and instantaneous reward γ . These elements form the tuple $(\mathcal{S}, \mathcal{A}, P, \gamma)$. Thus, a suitable MDP model is necessary to effectively handle the formulated problem. Based on the previous discussion, an MDP model customized for the optimization problem specified in Eq. (14) is constructed. Through iterative training, the optimal policy is obtained to maximize the expected reward.

In MDP model, the state represents an observation of the current environment, upon which the agent makes decisions. The future state depends solely on the current one, adhering to the memoryless property. In this research, the state is defined based on the information of each application queue in the gNB and the GCL settings for the time-triggered queue in the DS-TT. Specifically, the state space is defined as:

$$\mathcal{S} = \{RB_{exp}, L, D_{nor}, G\} \quad (15)$$

where each element of $\mathcal{S}$ is a vector. RB_{exp} represents the number of RBs required for each queue and is calculated by Eq. (9). L denotes the queueing packet number. G denotes the GCL state in DS-TT for time-triggered application. $G = 1$ means the GCL state of time-triggered queue in DS-TT is "OPEN", and $G = 0$ means "CLOSED". D_{nor} indicates the normalized queueing delay, each of which is calculated as:

$$d_{normalized,i} = \frac{d_i}{d_{qos}} \quad (16)$$

Based on the defined state space, the action space needs to be determined to complete the decision-making framework. As mentioned earlier, if the agent's output action is the number of RBs scheduled for each flow, it may lead to a high-dimensional action space. To address this issue, the action space in DDPG is confined to binary values:

$$\mathcal{A} = \{p_i\}_{(M+N)+1} \quad p_i \in \{0, 1\}, i \in F \quad (17)$$

where a value of "1" indicates that the corresponding flow will be scheduled in the current TTI, whereas a value of "0" indicates otherwise. Consequently, the RL model only needs to determine whether or not to schedule each flow. For the queues determined to be allocated resources, how to calculate the specific number of RBs to be allocated will be explained in the next section.

The reward function is designed to quantitatively describe the optimization goal for RL training, guiding the model to learn a better policy. It is designed according to the optimization target in Eq. (14) to provide guarantees for all flows in the 5G-TSN architecture and improve the overall system throughput, which is influenced by CQI. It is expressed as:

$$R(\varphi_i) = \begin{cases} \left(1 + \frac{d_i}{d_{qos}}\right) * \varphi_i & d_i \leq d_{threshold}, i \in F_t \cap G = 1 \\ -\left(1 + \frac{d_i}{d_{qos}}\right) * \varphi_i & d_i > d_{threshold}, i \in F_t \cap G = 1 \\ \left(1 - \frac{d_i}{d_{qos}}\right) * \varphi_i & d_i \leq d_{threshold}, i \in F_t \cap G = 0 \\ -\left(1 - \frac{d_i}{d_{qos}}\right) * \varphi_i & d_i > d_{threshold}, i \in F_t \cap G = 0 \\ 2 * \varphi_i & \forall i \in F_v \end{cases} \quad (18)$$

where φ_i is defined as:

$$\varphi_i = \begin{cases} \delta * \bar{th}_i & \bar{th}_i \geq th_{min}, i \in F_v \text{ or } \forall i \in F_t \\ 0 & \bar{th}_i < th_{min}, i \in F_v \end{cases} \quad (19)$$

Eq. (18) expresses the requirement for F_t . Considering the reliability of the d_i , its upper bound of it is restricted to $d_{threshold}$, which is lower than d_{qos} in F_t . Exceeding this threshold will incur penalties. Moreover, considering the characteristic of TSN domain, rewards increase when the GCL is open and decrease when it is closed. The magnitude of the reward increases or decreases is positively correlated with d_i . In Eq. (19), δ is implemented to control the deviation between different φ_i , aiming to keep the range of different reward values within 1000. $\bar{th}_i$ is required to be higher than th_{min} in F_v , otherwise, the reward is set to zero.

For each reward value of F_t , the coefficient is always lower than 2. However, for flows in F_v that $\bar{th}_i \geq th_{min}$, the coefficient δ is directly set to 2. Hence, the reward value for F_v is slightly higher than that for F_t . This design considers that F_v have larger packet size, thereby encouraging the scheduling of F_v while solidly ensuring the delay guarantee of F_t .

4.2. MCS-Embedded priority scheme

The DDPG algorithm makes 0–1 decisions to reduce the computational space for determining the specific flows to be scheduled in this part. According to Eq. (5), $\bar{th}_i$ is calculated based on CQI. According to Eqs. (13) and (15), maximizing $\bar{th}_i$ is a component of the optimization objective, which influences the allocation strategies. Thus, CQI is significant in this optimization model. CQI is influenced by various factors, including multi-path fading, multi-path interference, noise, etc. These factors show temporal and spatial variations, and there are correlations among them. Therefore, CQI is regarded as a non-Markov state and cannot be directly included in the MDP state space of DRL. In Table 1, CQI is mapped to the number of bytes per RB with related MCS b_i^f . In this section, MCS levels are utilized as the first priority to maximize the throughput and reduce the computation by replacing the DDPG's continuous action space.

Many data flows await scheduling in various users' queue due to the finite availability of RBs. In such circumstances, the transmission of F_t whose d_i is approaching their QoS should be prioritized. $d_{threshold}$ is used to distinguish urgent F_t and non-urgent F_t to ensure the latency reliability. Head-of-line (HOL) timestamps denote the generation time of the data, by examining them in all F_t queues, flows are classified based on their urgency. Specifically, for the flows that satisfy Eq. (20), then they will be deemed as urgent F_t :

$$t_{current} - t_{timestamp,i} \geq d_{threshold} \quad i \in F_t \quad (20)$$

where $t_{current}$ is the current time in system, $t_{timestamp,i}$ is the timestamp of flow i which belongs to F_t . Conversely, for flows whose differences of $t_{current}$ and $t_{timestamp,i}$ are less than $d_{threshold}$, they are considered non-urgent F_t .

The sum of RBs required by the urgent F_t , namely $RB_{urgent F_t}$. It is compared with $RB_{available}$ in the environment. If the RBs in environment

are insufficient, all RBs will be allocated to urgent F_t in order of MCS, and a warning of resource insufficiency will be published. On the contrary, if $RB_{available}$ are sufficient, the remaining RBs are then allocated to F_v in descending order of MCS. If the MCS levels of a part of data flows are identical, RBs are allocated based on their queue length.

The packet size of F_v is larger than that of F_t , as indicated by both the objective function Eq. (12) and the reward function of the DDPG algorithm Eq. (19). Therefore, after allocating RBs to urgent F_t , transmitting as many packets of F_v as possible is necessary. This process is represented as:

$$n_{F_v} = \left\lfloor \frac{RB_{available} - RB_{urgent F_t}}{RB_{F_v}} \right\rfloor \quad (21)$$

$$RB_r = (RB_{available} - RB_{urgent F_t}) \bmod RB_{F_v} \quad (22)$$

where n_{F_v} represents the number of F_v that can be transmitted, RB_{F_v} is the number of RBs required for each flow of F_v , and RB_r is the number of residual RBs. Then, the number of bytes corresponding to RB_r is compared with th_{min} . If it is less than th_{min} , RB_r is allocated to transmitting non-urgent F_t :

$$n_{non-urgent F_t} = \left\lfloor \frac{RB_r}{RB_{F_t}} \right\rfloor \quad (23)$$

where $n_{non-urgent F_t}$ is the packet number of non-urgent F_t that can be scheduled.

Otherwise, RB_r is used to partially transmit an additional F_v packet, with the unfinished part of this packet being kept in the scheduling queue for future transmission. For non-urgent F_t , resources are allocated first based on MCS levels, and then on queue length. If there are remaining RBs that are insufficient to meet the transmission requirements of a packet of F_t , these RBs are not utilized. After determining the n_{F_v} and $n_{non-urgent F_t}$, the packets to be scheduled are determined based on the priority mechanism, and the $RB_{scheduled,i}^t$ for each queue is calculated accordingly.

Eqs. (18) and (19) show that the reward function reflects the objective of maximizing throughput. To facilitates the DDPG algorithm in iterating towards a more optimal solution, this priority scheme is employed to ensure the QoS for different types of data flows. Since throughput enhancement cannot be observed in current episode, so the average throughput is maximized in each TTI.

In 5 G OFDM, an example of RBs allocation sequence associated with MCS-based priority scheme is shown in Fig. 4. Assuming each flow of F_t requires 1 mini-block, each flow of F_v requires 15 mini-blocks, th_{min} of each flow in F_v requires 10 mini-blocks. This model first allocates resources to urgent F_t to ensure the delay guarantee. Then it schedules the flows of F_v , but the available RBs only support the transmission of 3 flows. It is assumed that the unscheduled F_v contains one high-MCS flow and one medium-MCS flow, so a low-MCS flow will be scheduled. After determination, 8 mini-blocks remain, then the non-urgent flows of F_t are allocated to them. But if the number of remained blocks meets the th_{min} of F_v , a part of flow in F_v will be allocated to them.

Overall, the 5G-TSN joint resource scheduling action is decomposed into two components, one is executed through DDPG, and the other is handled by the MCS-based priority method. The whole resource scheduling process is detailed in Algorithm 1. Through iteration, the agent can select the channel with better CQI to allocate the limited resources, thereby improving various aspects of performance.

Algorithm 1. Resource scheduling process based on the ME-DDPG

1. Initialize replay memory
2. Initialize the channel model and DDPG model
3. for episode=1,2,...E do
4. Initialize the packet model
5. for $t = 1, 2, \dots$ Step do
6. Given the current state S^t .

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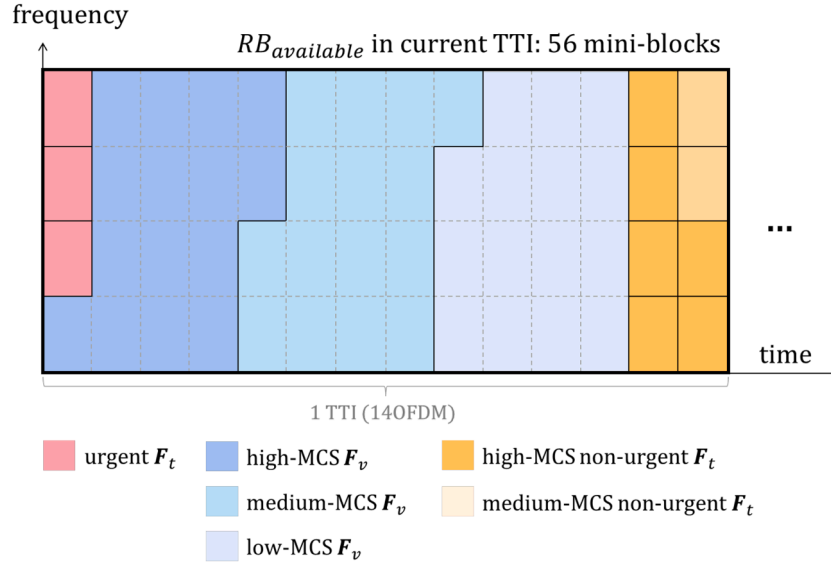


Fig. 4. An Example of Allocation Sequence of Specific RBs.

(continued)

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7.   Get binary action  $A^t$  from  $\pi(S^t; \Theta)$ . // binary outputs determine which flows
   are scheduled or not
8.   if Eq. (10) is satisfied then
9.      $RB_{scheduled, i}^t \leftarrow RB_{required, i}^t$ 
10.  else
11.    for each queue in the unscheduled queues of  $F_t$  do // unscheduled queues are
   those where  $A_i^t = 1$ 
12.      Determine urgent  $F_t$  and non-urgent  $F_t$  using Eq. (21).
13.    end for
14.    Calculate the  $RB_{urgent, F_t}$ .
15.    if  $RB_{available} - RB_{urgent, F_t} < 0$  then
16.      Determine urgent  $F_t$  to be scheduled based on two-tier priority scheme
   (MCS level and queue length)
17.      Publish a warning of resource insufficiency.
18.    else
19.      All urgent  $F_t$  are considered for immediate scheduling.
20.      Calculate  $n_{F_t}$  and  $RB_r$  using Eq. (22).
21.      Determine  $F_v$  to be scheduled based on two-tier priority scheme.
22.    if  $RB_r > RB_{th_{min}}$  then
23.       $RB_r$  used to transmit a part of additional  $F_v$ , the remainder kept in queue.
24.    else
25.       $RB_r$  used to schedule the non-urgent  $F_t$  using Eq. (23) to attain  $n_{non-urgent, F_t}$ .
26.      Determine non-urgent  $F_t$  to be scheduled based on two-tier priority scheme.
27.    end if
28.  end if
29.  Calculate  $RB_{scheduled, i}^t$  in each queue based on the determined flows.
30.  end if
31.  Follow the  $RB_{scheduled, i}^t$  to allocate RBs to each flow  $i$  and get reward  $R^t(\varphi_i)$  with
   Eq. (18).
32.  Transit to the next state  $S^{t+1}$ .
33.  Store  $(S^t, A^t, R^t(\varphi_i), S^{t+1})$ .
34.  Update the DDPG network.
35. end for
36. end for

```

Table 2

Simulation parameters.

Simulation parameters	Value
The probability of packets arriving within each TTI	0.8
The number of packets arriving within each TTI	The number is valued from 1 to 3, followed by a uniform distribution
Packet size of time-triggered flow	50 Bytes
Packet size of Video Flow	1000 Bytes
gNodeB Transmit power	20 dBm
Channel Noise power	-90 dBm
Subcarrier interval	15 kHz
d_{qos}	10ms
$d_{threshold}$	8.5ms
th_{min}	400 Bytes
α	0.01
β	0.01
δ	0.001
Discount Factor γ	0.9
Episode number	500
TTI number in each episode	500
Batch Size	32

Rayleigh distribution with a standard deviation of 1.0. The distance between the gNB and UE is subject to a uniform distribution ranging from 0 to 1 [34]. Additionally, the transmission power of gNB is 20 dBm, and the noise power of the channel is -90 dBm. The sub-carrier spacing set in this study is 15 kHz, which corresponds to a TTI period of 1 ms. The d_{qos} of F_t is 10ms. In the reward function in Eq. (18), the delay threshold of F_t is $d_{threshold}$, and exceeding this threshold results in a penalty to the reward value. And according to Eq. (2) and Eq. (19), th_{min} of each flow in F_v is set to 400 Bytes. α and β represent the learning rates of the actor network and the critic network. γ is the discount factor. In whole model training, we have set 500 episodes and each episode has 500 TTI.

Table 3 presents the architecture of ME-DDPG. The Actor network includes an input layer with a fully connected structure of $N_{UE} \times 4$ and ReLU activation function, followed by a normalization layer; a hidden layer with $N_{UE} \times 4 \times 5$, and ReLU activation function, also followed by a normalization layer; and an output layer with $N_{UE} \times 1$ using a Tanh activation function. The Critic network consists of an input layer with $N_{UE} \times 5$ and ReLU activation function, followed by a normalization layer; a hidden layer with $N_{UE} \times 5 \times 5$ and ReLU activation function, also followed by a normalization layer; and an output layer with a single

5. Simulation results and discussion

5.1. Simulation parameters

Table 2 presents the simulation parameters. The UE model follows a Bernoulli distribution with a probability of 0.8 for generating new data packets in each TTI. According to 3GPP TS 22.104, time-triggered applications generate smaller data packets, while video applications generally produce larger ones. Therefore, the packet size in F_t is set to 50 bytes and that in F_v is set to 1000 bytes. The channel model follows a

Table 3
Network parameters.

Layers	Actor network	Critic network
Input layer	Fully connected layer ($N_{UE} \times 4$) ReLU Nonlinear activation function Normalization Layer	Fully connected layer ($N_{UE} \times 5$)
Hidden layer	Fully connected layer ($N_{UE} \times 4 \times 5$) ReLU Nonlinear activation function Normalization Layer	Fully connected layer ($N_{UE} \times 5 \times 5$)
Output layer	Fully connected layer ($N_{UE} \times 1$) Tanh Nonlinear activation function	Fully connected layer (1)

unit.

To evaluate the impact of different GCL settings in DS-TT on 5 G scheduling, three distinct GCL schemes are compared. The first scheme alternates the time-triggered queue state between “ON” for one TTI and “CLOSE” for the next, represented as “0101”. In the second scheme, the GCL state changes every two TTIs, denoted as “0011”. The third scheme selects the “ON” state with a probability of 0.5, which is named as “random.”

In this study, six user configurations are considered for simulation for joint resource scheduling, including 2 time-triggered users with 7 video users ($|F_t|: |F_v| = 2: 7 \approx 0.286$), 3 time-triggered users with 9 video users ($|F_t|: |F_v| = 3: 9 \approx 0.333$), 5 time-triggered users with 10 video users ($|F_t|: |F_v| = 5: 10 = 0.5$), 7 time-triggered users with 11 video users ($|F_t|: |F_v| = 7: 11 \approx 0.636$), 9 time-triggered users with 12 video users ($|F_t|: |F_v| = 9: 12 = 0.75$) and 12 time-triggered users with 12 video users ($|F_t|: |F_v| = 12: 12 = 1$). As the total number of users increases, the proportion of time-triggered flow increases.

The network simulation generates packets based on the number of users, which affects the experimental results. Fig.5 illustrates the number of packets per episode under two configurations: the minimum number of users ($|F_t|: |F_v| = 2: 7$) and the maximum number of users ($|F_t|: |F_v| = 12: 12$). Through calculation, the standard deviation of packets per episode in the minimum-user scenario remains within 10 % of the average, indicating steady performance. Similarly, in the maximum-user case, the deviation remains within 5 %, which is further stable.

To ensure fairness and consistency in algorithm comparisons, the same random seed was used for packet generation after achieving stability in packet counts. This approach eliminates errors caused by the randomness and ensures that all algorithms are compared under identical testing conditions, making the evaluation results more reliable and reproducible.

5.2. Simulation results

To evaluate the performance of different algorithms and configurations under diverse scenarios, extensive simulations were conducted. The results are presented and analyzed across multiple dimensions, including reward stability, latency performance, reliability, and throughput.

The DDPG-based algorithm proposed in [31] shows that this algorithm better guarantees the d_{qos} of F_t and promotes the throughput of F_v , while also exhibiting greater universality. This algorithm employs the ordinary DDPG algorithm (hereafter referred to as O-DDPG), where the reward function is designed to postpone the time-triggered flows from approaching their QoS limits as much as possible, thereby promoting the transmission of video flows. After the DDPG algorithm makes binary

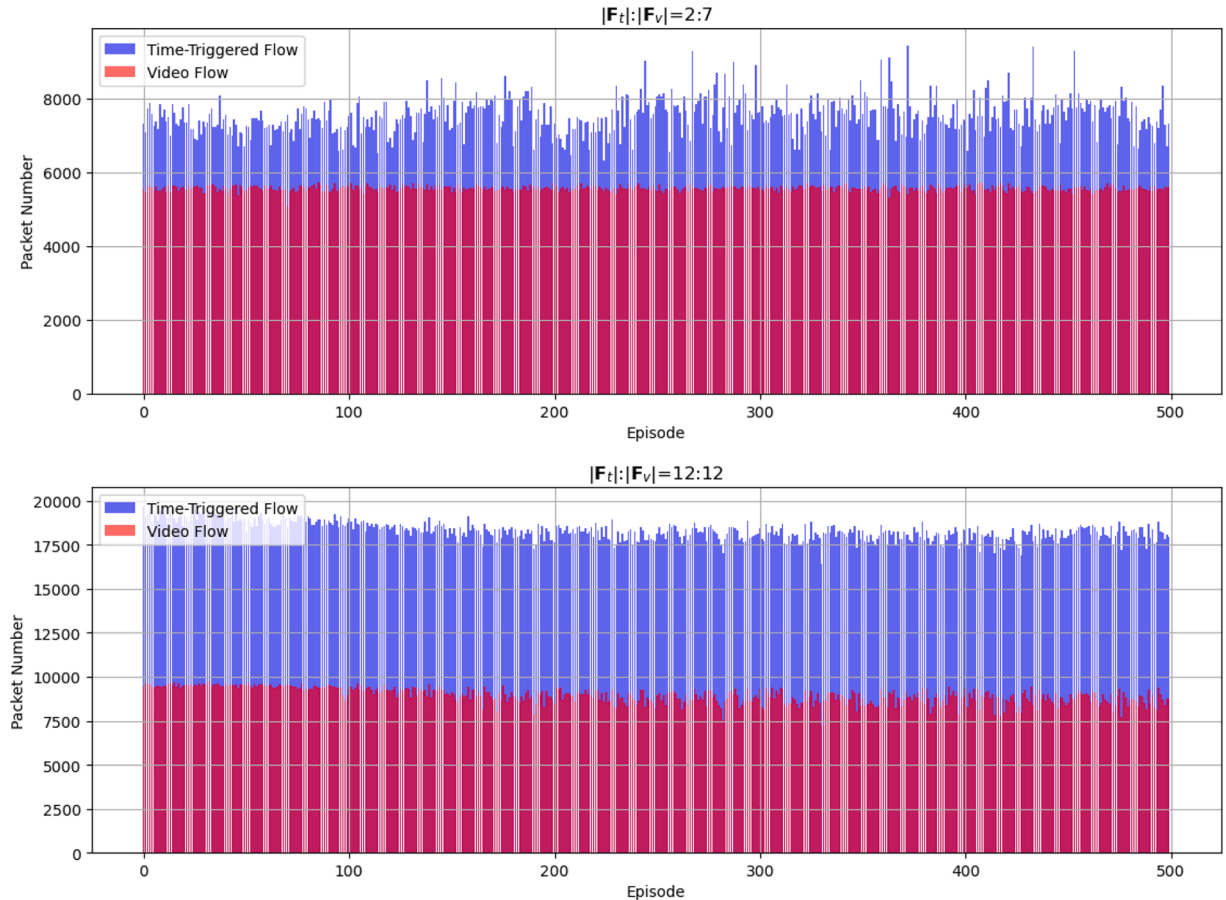


Fig. 5. Packet Number Vs Episode.

decisions for resource allocation, specific assignments are simply made according to the proportion of the RB's number required in each queue. However, this approach does not consider the impact of channel quality and the potential negative effect on the reliability of time-triggered flows due to the postpone. Compared with the O-DDPG, ME-DDPG has fewer neural network layers for two-dimensional action output and modifies the reward function.

The DPF-based algorithm proposed in [32] addresses the resource scheduling problem in URLLC scenarios by integrating a pricing mechanism and non-cooperative game theory. By incorporating delay constraints into the utility function, this algorithm effectively balances transmission reliability, scheduling delay, and fairness, while dynamically maximizing throughput. However, its reliance on static traffic priorities may fail to adapt to dynamic channel variations, potentially limiting the throughput of lower-priority flows and impacting overall performance in scenarios with mixed traffic types.

5.2.1. Reward convergence and stability

Since both the O-DDPG and ME-DDPG are RL algorithms, they first made a comparison in terms of the reward value.

Fig.6 depicts the normalized reward convergence of O-DDPG and ME-DDPG under different user configurations, with the GCL set to “random” for both algorithms. Under all user proportions in Fig.6, the reward of the proposed algorithm can converge more smoothly than that of the O-DDPG. It reflects that the reward function design in this article is more reasonable, and the priority consideration is suitable for reward function. Concretely, the MCS-based strategy takes overall throughput and into account and divides F_t into urgent flows and non-urgent flows, allowing it to make more informed decisions during training. Moreover, a smoothly converging reward helps to reduce significant fluctuations in the policy update process, making the model iteration process more reliable. To quantitatively describe the reward value, the average reward, convergence speed, and stable reward are calculated in Table 4. The methods for calculating these metrics are described as follows:

Average Reward: The average reward is computed by taking the

Table 4
Comparison of Reward Value.

User Configuration	Algorithm	Average Reward	Convergence Speed (step)	Stable Reward
$ F_t : F_v =2:7$	O-DDPG	0.6407	118	0.6857
	ME-DDPG	0.8877	85	0.9882
$ F_t : F_v =3:9$	O-DDPG	0.7467	49	0.7959
	ME-DDPG	0.8796	92	0.9873
$ F_t : F_v =5:10$	O-DDPG	0.7629	79	0.8135
	ME-DDPG	0.8818	90	0.9882
$ F_t : F_v =7:11$	O-DDPG	0.6966	195	0.7565
	ME-DDPG	0.8801	89	0.9872
$ F_t : F_v =9:12$	O-DDPG	0.7403	146	0.7441
	ME-DDPG	0.8812	98	0.9908
$ F_t : F_v =12:12$	O-DDPG	0.8043	48	0.8092
	ME-DDPG	0.8798	98	0.9877

mean of all reward values over 500 episodes. This provides an overall measure of the reward performance. Our algorithm attains a higher average reward in all user configurations.

Convergence Speed: The convergence speed is determined by identifying the number of episodes for the reward values to reach a stable level, calculated as the point where they first stabilize within a predefined threshold (in this case, 0.8) and remain stable for subsequent iterations. Although the O-DDPG sometimes converges faster, in the cases where it converges more slowly, it requires approximately twice the number of episodes as the proposed algorithm.

Stable Reward: The stable reward is calculated as the mean reward value after the convergence point has been reached. This metric indicates the performance of the system once it has stabilized and can be considered a measure of long-term performance. In this article, the average reward values in the last 100 episodes are chosen as the convergence rewards. As shown in Table 4, the reward of proposed algorithm is indicated greater stability.

In Table 4, it can be observed that the algorithm proposed in this

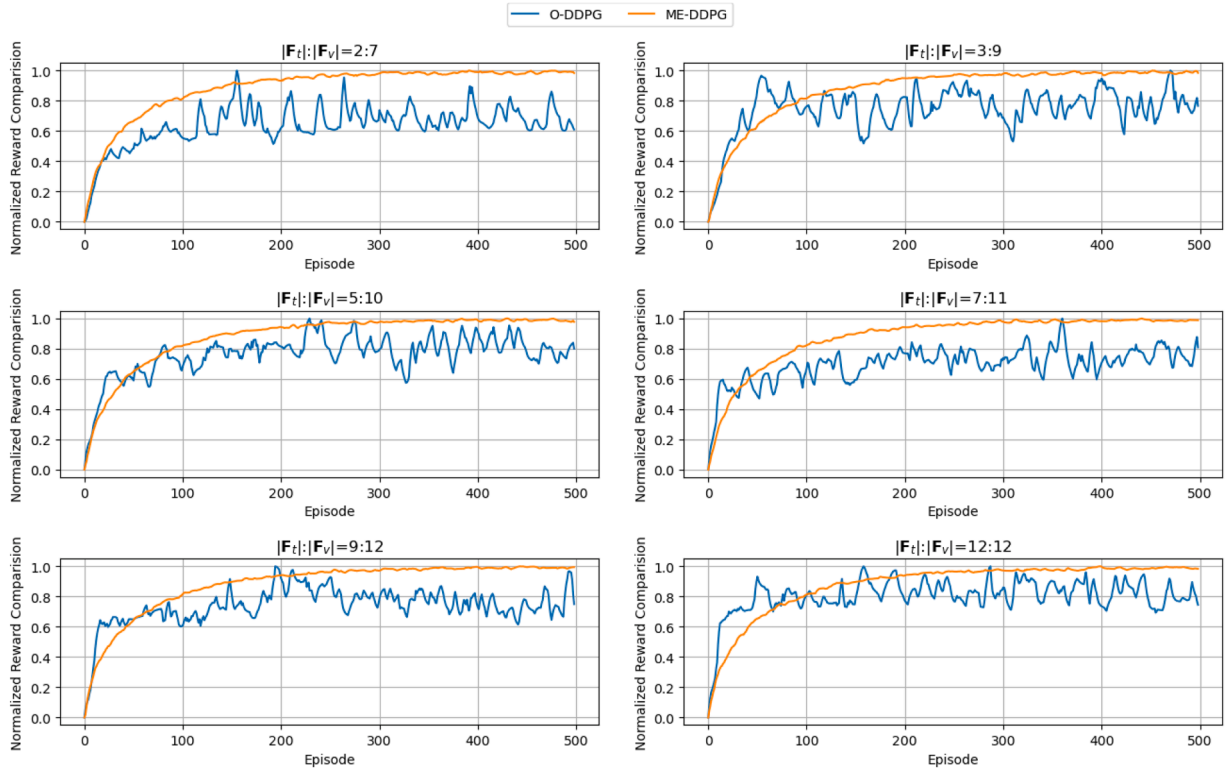


Fig. 6. Reward Comparison Vs Episode.

article outperforms the O-DDPG in terms of reward values. To further analyze its performance, the reward values of the proposed algorithm is compared under different GCL settings, as depicted in Fig. 7. Specifically, the standard deviation of the reward values during the convergence process under different GCL settings is relatively small, which reflects its robustness in handling different network conditions. In RL, the stabilization and near-convergence of the reward function suggest that the agent has gradually found a relatively stable strategy, enabling it to continuously receive favorable feedback in a given environment. This implies that the agent's understanding of the environment and its action strategy has reached a relatively mature stage.

5.2.2. Latency performance

In our models, time-sensitive services are not required to have the lowest possible latency, as long as they remain within the acceptable QoS range. Fig. 8 illustrates the comparison of average d_i of F_t under "random" GCL setting. It demonstrates that all the three algorithms can guarantee the QoS of F_t to a certain extent, but the ME-DDPG have an edge in terms of latency optimization. It utilizes a more sophisticated mechanism for resource allocation and scheduling, which involves the real-time analysis of conditions within both 5 G domain and the TSN domain, thereby making more precise decisions on latency control.

Furthermore, as depicted in Fig. 9, it is evident that under diverse GCL settings in the TSN domain, ME-DDPG consistently maintains the latency within the QoS requirements. The algorithm leverages real-time analysis of network conditions and applies dynamic adjustments to prioritize urgent traffic flows. Our approach not only guarantees the required QoS but also suits well to varying network conditions, ensuring reliable performance in a wide range of scenarios.

5.2.3. Reliability performance

Reliability is defined as the proportion of d_i below to d_{qos} in all

episodes and is plotted in Fig. 10. It is used to compare the latency performance under different GCL settings and user configurations. The ME-DDPG selects channels with better CQI in resource-constrained episodes, achieving 100 % reliability in all scenarios and episodes. In contrast, DPF and O-DDPG only occasionally achieves 100 % reliability. This result indicates that the proposed ME-DDPG can better guarantee the delay of F_t , which is conducive to 5G-TSN transmission environments.

5.2.4. Throughput performance

Fig. 11 illustrates the comparison of the average throughput under different user ratios and GCL settings. Each throughput value is calculated as the mean value over 500 episodes. When compared with O-DDPG, the ME-DDPG enhances overall throughput at least 9.16 %, and when compared with DPF, the ME-DDPG achieves an overall throughput improvement in the range of 10.84 %. Among them, the throughput of F_v and that of F_t have both been significantly improved. Due to the average throughput is positively correlated with the increasing proportion of F_v in ME-DDPG. It is indicated that considering CQI and utilizing it in a priority scheme to resource allocation positively impacts throughput improvement, enabling more video flows to be transmitted. Furthermore, the algorithm consistently improves throughput performance under different GCL settings. In IIoT scenarios, where a larger amount of data needs to be transmitted, thus, the enhancement of the throughput transmission contributes to stronger network performance.

6. Conclusion and future works

This study addresses the critical challenge of joint resource scheduling in 5G-TSN networks, particularly in IIoT scenarios requiring deterministic transmission and efficient resource utilization. The proposed ME-DDPG algorithm combines a DDPG model with an MCS-based

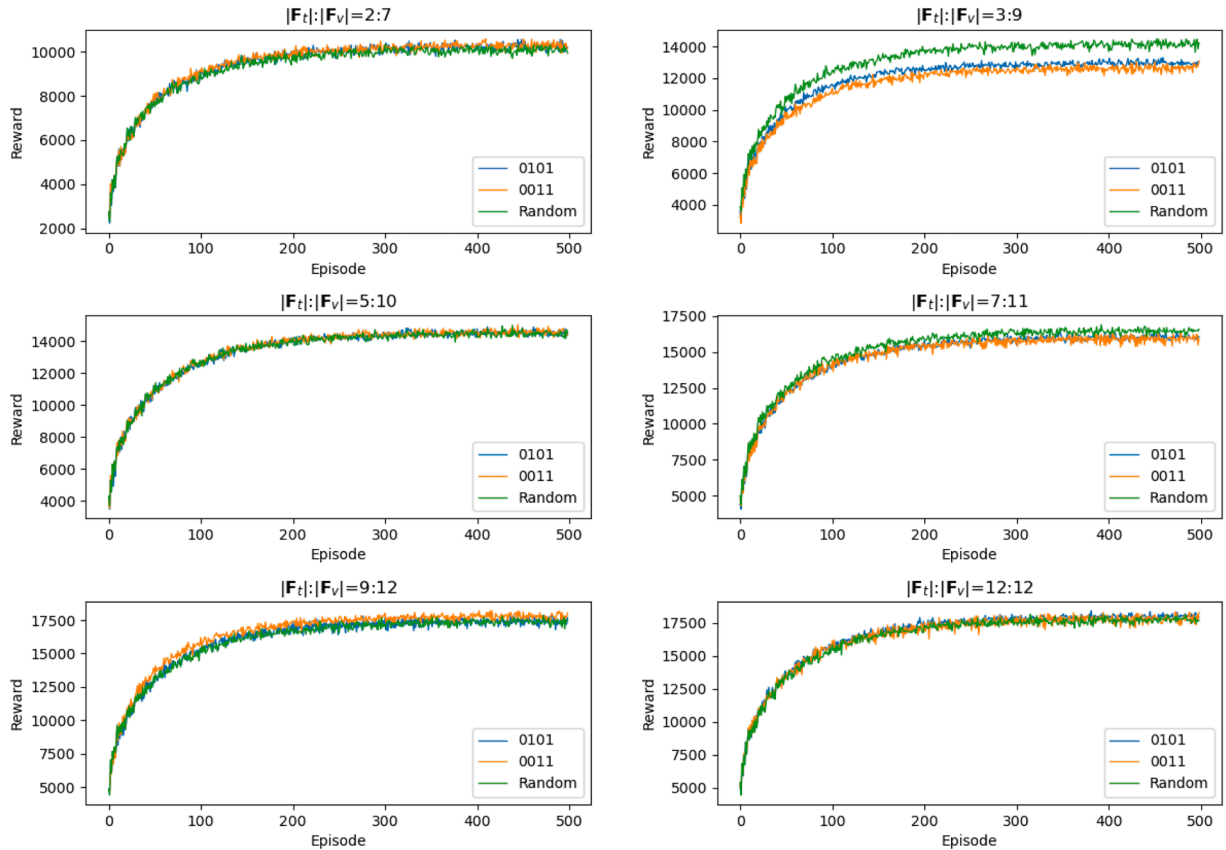


Fig. 7. Reward of the Proposed Algorithm under Different GCL Settings.

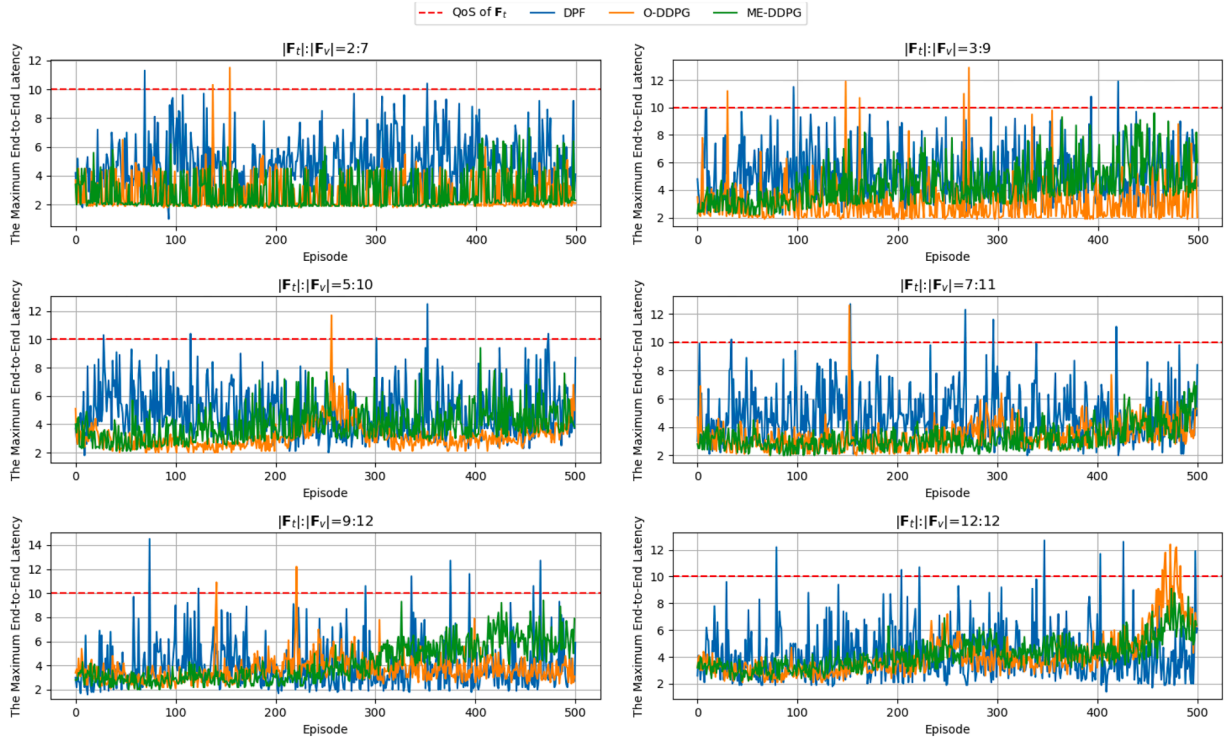


Fig. 8. Comparison of Latency Performance.

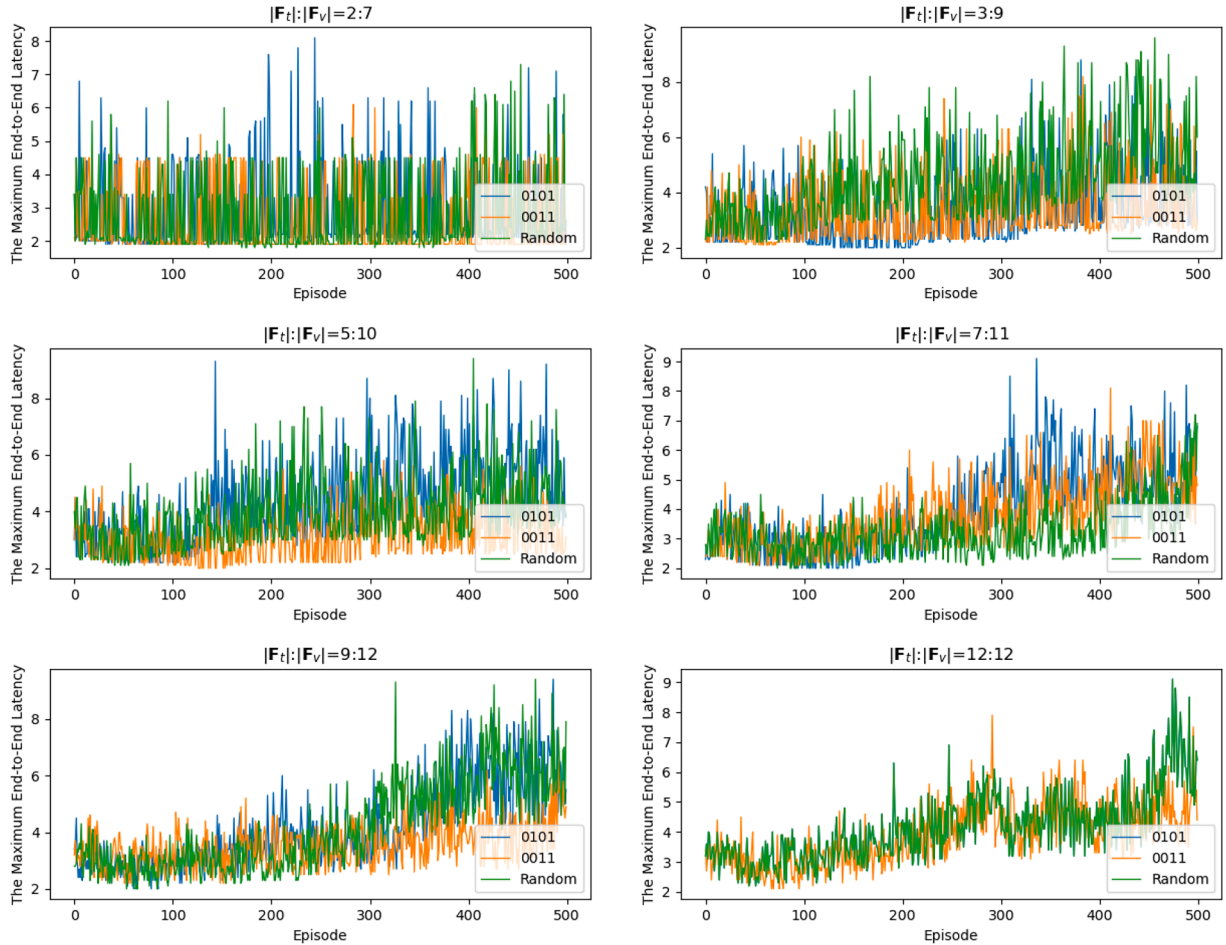


Fig. 9. Latency Comparison of Different GCL Settings.

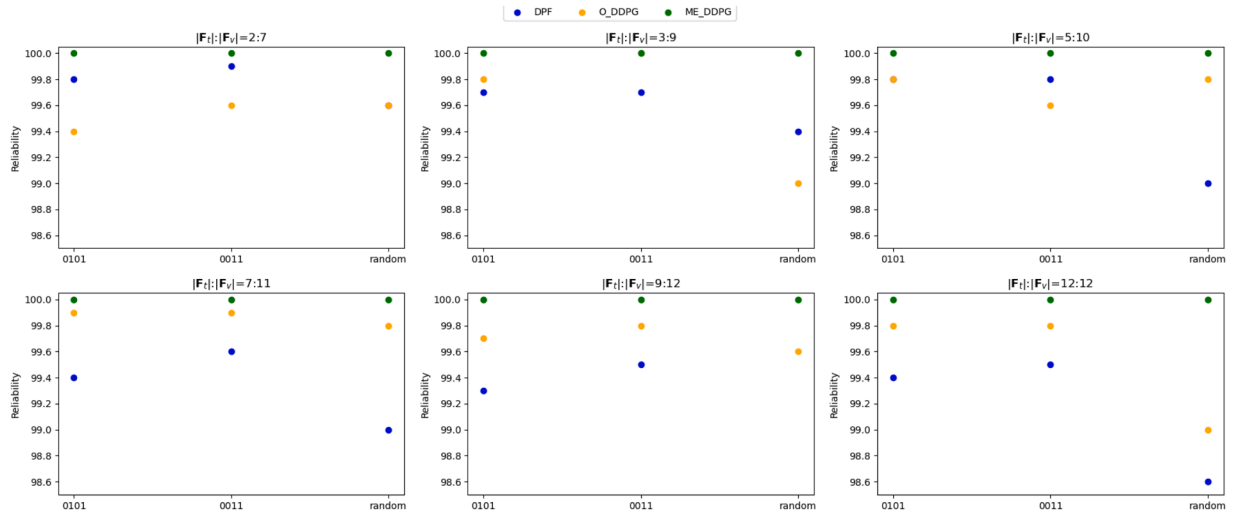


Fig. 10. Reliability Comparison.

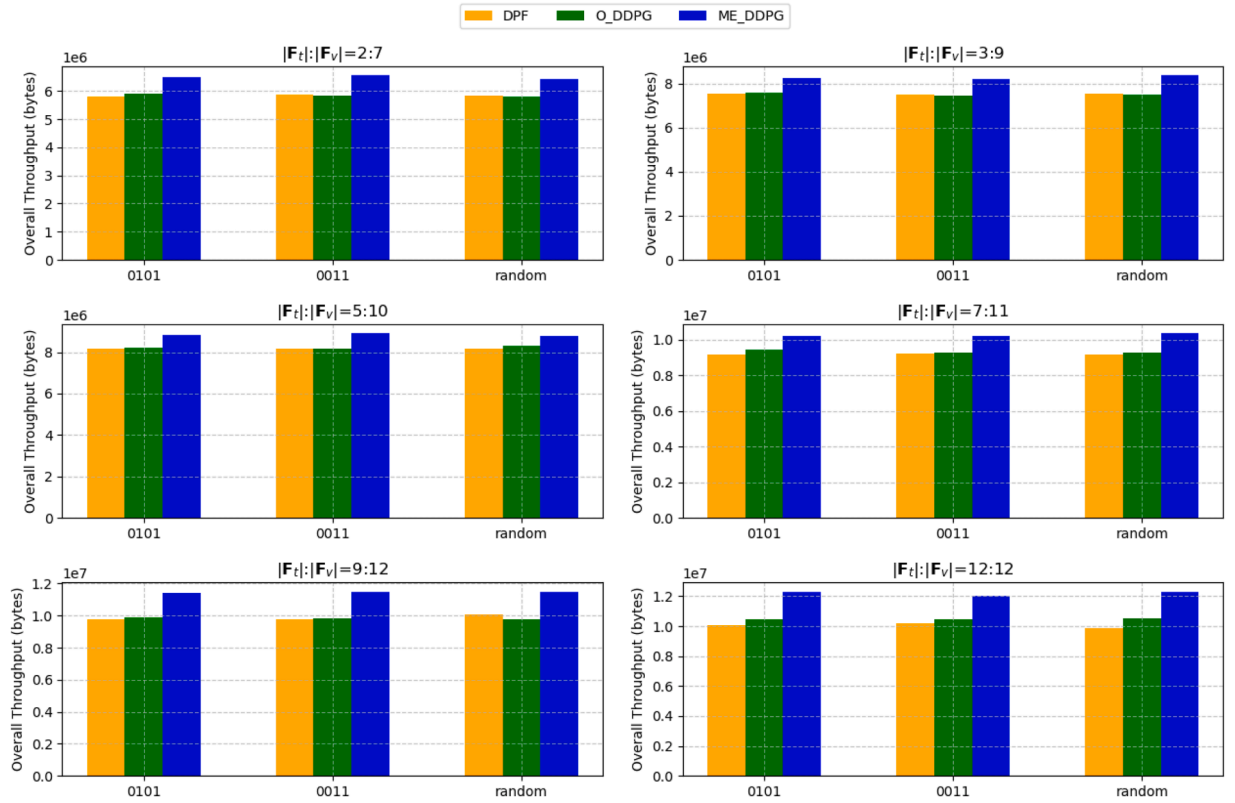


Fig. 11. Throughput Comparison.

priority scheme to optimize latency and overall system through radio resource allocation. By leveraging CQI to dynamically prioritize data flows and by reducing the action space, ME-DDPG achieves significant computational efficiency and adaptability, making it highly suited for complex industrial environments. Simulation results demonstrate that ME-DDPG can provide reliable latency guarantee for time-triggered flows under various configurations, ensuring robust QoS for time-sensitive applications. Additionally, ME-DDPG deployed an MCS-based resource allocation optimization model within DDPG, which accelerates reward function convergence and improves overall system throughput compared to baseline methods, effectively addressing the bandwidth-intensive requirements of video flows. These research

outcomes highlight the capability of ME-DDPG to balance multiple QoS objectives. This is particularly essential for large-scale deployment in industrial scenarios, where diverse applications with various QoS requirements coexist and should be satisfied.

While ME-DDPG demonstrates strong performance, further work is needed to expand its applicability to more industrial scenarios. Applications such as motion control, which require ultra-low and bounded jitter, present additional challenges. Even slight timing deviations in these contexts may lead to task failures or degraded product quality. Future research will focus on developing distributed decision-making mechanisms among multiple agents to enable real-time synchronization, improving jitter control while maintaining low latency and high

throughput. In those scenarios, motion control scenarios can be addressed by designating motion devices as DRL agents, synchronizing their local state information, and adapting their reward function based on dynamic 5 G resources availability and GCL configuration in TSN. Additionally, deploying ME-DDPG in real-world industrial environments will provide valuable insights into its practical performance and validate its feasibility for large-scale IIoT applications. By addressing these future directions, the ME-DDPG framework can further establish itself as a versatile and robust solution for next-generation industrial communication networks.

CRedit authorship contribution statement

Yang Zhang: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Lei Sun:** Writing – review & editing, Validation, Supervision, Funding acquisition. **Zhangchao Ma:** Writing – review & editing, Validation, Supervision. **Jianquan Wang:** Writing – review & editing, Validation, Supervision, Funding acquisition. **Meixia Fu:** Writing – review & editing, Validation, Supervision. **Jinoo Jung:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Jianquan Wang reports financial support was provided by Natural Science Foundation of Beijing Municipality with Grant No. L232001. Lei Sun reports financial support was provided by Beijing Science and Technology Planning Project with Grant No. Z231100005923025. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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